UGV-UAV Cooperative 3D Multi-Object Tracking Based on Multi-source Data Fusion

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***Abstract*—Unmanned aerial vehicles (UAVs) and Unmanned ground vehicles (UGVs) have complementary sensing characteristics, and their cooperative sensing can effectively solve the problems of target loss due to view occlusion or distance in multi-target tracking tasks. In this paper, we propose a track by detection framework based on a multi-source data fusion approach. In the detection phase, a modified ImvoxelNet network with attention mechanism is proposed to fuse the multi-view image data of unmanned vehicles and UAVs to estimate the position and size of the target. In the tracking stage, the method involves correlation processing of both 2D and 3D detection data. It utilizes an interactive multi-mode unscented Kalman filter (IMM-UKF) algorithm that incorporates the Minimum-update Successive Shortest Path (muSSP) method to enhance the accuracy and real-time performance of dynamic target tracking. The experimental results demonstrate the effectiveness of the proposed approach in significantly improving the performance of multi-target tracking tasks.**

***Keywords*—UGV-UAV, Cooperative Perception, 3D Multi-Object Tracking, Multi-source Data Fusion, IMM-UKF**

I. INTRODUCTION

3D multi-object tracking (MOT) constitutes a crucial component in various applications, such as autonomous driving and assistive robotics [1]. Typically, MOT methods adhere to the tracking by detection paradigm, comprising two distinct steps: detection and data association. During the detection step, potential box predictions of target objects are generated for each frame. Subsequently, data association step aims to match the predicted boxes of the same target across frames, leveraging appearance and motion cues. There have been numerous methods proposed in recent years, resulting in significant improvements in detection accuracy. However, despite these advancements, a single unmanned vehicle still faces challenges in efficiently accomplishing target tracking tasks, owing to the inherent limitations of the camera's perception view and the LiDAR's perception distance.

Cooperative perception methods offer several significant advantages, including providing a global perspective that extends beyond the current horizon and covering blind spots. When compared with V2V and V2X communication methods, UAVs possess characteristics of flexibility and deployability, making them capable of mounting cameras to acquire low-altitude, high-resolution image information [2]. This feature finds extensive application in the target detection and tracking domain. Fig. 1 showcases a multi-objective tracking task scenario that employs a UGV-UAV cooperative sensing system to track vehicles obscuring pedestrians. To address challenges like full occlusion and object misses due to temporary occlusion by other objects or being distant from the LiDAR sensor, multi-modality MOT methods are utilized to preserve the object's ID.

Fig. 1. An illustration of multi-target tracking task using the UGV-UAV cooperative sensing system in a scenario where pedestrians are occluded by vehicles.

This paper presents a progressive framework for track-by-detection based on a multi-source data fusion method. During the detection phase, we propose an ImvoxelNet network featuring an improved attention mechanism to fuse multi-view image data from UGV and UAV sources. This enables us to perform target detection, as well as positional and dimensional estimation of the targets. In the tracking stage, we adopt the AB3DMOT tracking method for the correlation and fusion processing of 2D and 3D detection data. Additionally, we utilize Minimum-update Successive Shortest Path method combined with the IMM-UKF algorithm to enhance the accuracy and real-time performance

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of dynamic target tracking. By leveraging multi-stage and multi-level data fusion processing, we effectively address the issue of frequent target ID switching in occlusion scenarios.

The subsequent sections in this paper are structured as follows: Section II provides a concise review of the latest advancements in the field. In Section III, a mathematical framework for the cooperative multi-object tracking problem is introduced, along with the presentation of the enhanced algorithm. Real-world traffic scenarios are utilized to conduct various experiments, and the outcomes are outlined in Section IV. The paper concludes with Section V, which offers a discussion on future research directions.

II. RELATED WORKS

Multi-modal fusion methods for Multi-Object Tracking (MOT) tasks in autonomous driving have seen significant advancements [3]. Various approaches in this domain encompass a wide spectrum, incorporating sophisticated cross-modal feature representations, enhanced sensors with improved reliability in different modalities, and intricate deep learning models and fusion methods. These techniques encompass a diverse range of fusion strategies, such as Early-fusion, Deep-fusion, Late-fusion, and Asymmetry-fusion, each designed to address specific challenges and optimize multi-modal data integration [4].

DeepSORT [5] incorporates a deep feature extractor to capture discriminative appearance features and applies a Kalman filter-based motion model for accurate state estimation. However, DeepSORT heavily relies on precise object detection, making it vulnerable to errors or inaccuracies in the detection stage. AB3DMOT [6] takes a step further and integrates 3D object detection and association for tracking objects in a three-dimensional space, making it particularly well-suited for 3D MOT using point cloud data. Meanwhile, DeepFusionMOT [7] aims to enhance MOT performance by fusing appearance and motion cues through a deep fusion network. It combines appearance features from image-based object detectors with motion features from optical flow or motion estimation algorithms. A key obstacle encountered by current MOT techniques lies in the successful integration of LiDAR-camera data, where the absence of depth information in camera images poses a challenge. To tackle occlusion issues, a novel approach is employed, involving the computation of depth information for occluded objects from 2D detections. By projecting both 2D and 3D detections onto the horizontal plane, this method effectively eliminates occlusion, revealing relative object locations when viewed from the bird's-eye view (BEV). As a result, the LiDAR-camera fusion becomes more accurate and robust in handling challenging scenarios.

ImVoxelNet [8] constructs a voxel representation of the 3D space to accumulate information and make predictions from multiple input images. Point painting and MVXNet [9] are approaches used to fuse image and LiDAR features for predicting 3D bounding boxes. DAIR-V2X introduces a late fusion framework specifically designed to tackle the Vehicle-Infrastructure Cooperative 3D Object Detection problem. Despite notable progress in 3D object detection, challenges persist, including issues related to blind spots and limited long-distance perception. SARPNET [10] presents a novel feature encoder that effectively deals with the sparsity and inhomogeneity present in point clouds. Moreover, it integrates an attention mechanism to learn the 3D shape characteristics of objects. In this work, we employ the attention mechanism to enhance the network's focus on areas of interest, thereby reducing the incidence of false negatives and false positives.

In order to improve the precision of data association, modern MOT frameworks often integrate a predictor component to predict the trajectory of targets. MOT can be classified into two main approaches: filter-based techniques and learning-based methods. Filter-based methods estimate the object state using a theoretical model, whereas learning-based approaches utilize time-sequence data to perform state estimation. Some tracking algorithms, such as references [11-12], utilize the classical Kalman filter (KF) for predicting track states in subsequent frames, providing improved real-time performance without the need for training. The Unscented Kalman filter (UKF) [13], known for its ease of implementation and accuracy, outperforms the Kalman filter. The IMM-UKF algorithm inherits the IMM algorithm's capacity to adapt to the modal modulation of motorized targets and the excellent accuracy of the UKF filter. In the literature, the IMM algorithm is combined with nonlinear filtering to fully utilize model information to weight the inputs and outputs of the filters for adaptive filtering.

The IMM algorithm's performance is influenced by the selection of the Markov state transfer matrix, model parameter noise, and target motion model. An improved radar and infrared sensor cooperative tracking algorithm is proposed for use in radar and infrared sensor cooperative tracking systems based on IMM-UKF [14]. Additionally, IMM-JPDA-STUKF [15] combines strong tracking filter (STF) theory with UKF to improve the filtering accuracy and robustness of UKF. Furthermore, [16] proposes a robust adaptive UKF (RAUKF) to improve state estimation accuracy and robustness with uncertain noise covariance. We tackles consecutive frame prediction problems by incorporating track uncertainty into the cost function design, leading to uncertainty-guided data association. The modified IMM-UKF algorithm is designed to effectively handle the inherent uncertainty associated with vehicle movement while demonstrating commendable real-time performance.

III. METHODS

A. Problem Statement

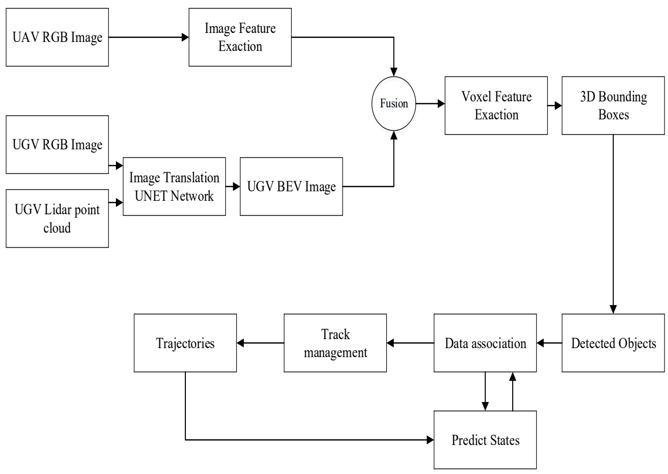

Fig. 2. The track by detection framework based on a multi-source data fusion approach.

Illustrated in Figure 2, we introduce an innovative architecture for 3D object detection and tracking that utilizes fusion-based techniques. Within the context of 3D MOT, IMM approach formulates the problem as tracking multiple objects in a 3D space over time. Each object is represented by a state vector comprising its position, velocity, and other relevant attributes. The key assumption of the IMM model is that the state dynamics of objects can switch between different motion models, each representing a distinct motion pattern or behavior. The central idea revolves around maintaining a set of Kalman filters, with each corresponding to a specific motion model, and then combining their estimates to achieve robust tracking outcomes.

$$X = [x, y, \theta, v, \omega]^{\mathrm{T}} \tag{1}$$

$$X_{t+1}^{CV} = \begin{bmatrix} x_t + v\delta_t \cos\theta \\ y_t + v\delta_t \sin\theta \\ \theta \\ v \\ 0 \end{bmatrix} \tag{2}$$

$$X_{t+1}^{CTRV} = \begin{bmatrix} x_t + \frac{v}{\omega}\sin(\theta + \omega\delta_t) - \frac{v}{\omega}\sin(\theta) \\ y_t + \frac{v}{\omega}\cos(\theta + \omega\delta_t) + \frac{v}{\omega}\cos(\theta) \\ \theta + \omega\delta_t \\ v \\ \omega \end{bmatrix} \tag{3}$$

$$\widetilde{X}_j = \sum_{i=1}^{n} w_{ij} X_i \tag{4}$$

$$\widetilde{P}_j = \sum_{i=1}^{n} w_{ij} \left[P_i + \left(X_i - \widetilde{X}_i\right) \cdot \left(X_i - \widetilde{X}_i\right)^{\mathrm{T}} \right] \tag{5}$$

Equation (1) is The state vector of a moving target contains its pose information and motion information. Equation (2) and (3) are the system functions of the CV and CTRV motion models. Equation (4) and (5) are state and covariance using model prediction probabilities and mixed probability representations. These probabilities are updated based on the likelihood of the measurements under each motion model. By continually updating and combining the estimates from different motion models, the IMM model demonstrates its capability to handle abrupt changes in motion patterns, occlusions, and other challenging scenarios that are typical in 3D MOT. As a result, it offers a versatile and effective approach for tracking multiple objects in complex and dynamic environments.

B. 3D Object Detection

In this study, we have introduced the Attention Mechanism module based on ImVoxelNet to enhance the network's focus on specific targets of interest, thereby directing more attention towards particular regions or objects. The integration of the Attention mechanism in vehicle multi-target tracking methods allows the utilization of Attention feature maps with weights to enhance the representation of vehicle targets in the feature maps.

Fig. 3 shows the proposed Attention Mechanism module. After input the features, the Spatial Attention (SA) module first performs Max pooling and Average pooling based on the number of feature channels. These pooled features are then combined through a channel-wise union, resulting in a 1-dimensional output with one channel, achieved by a convolutional layer that downsizes the input feature map. By incorporating the channel attention mechanism, the network can effectively emphasize important features and improve the extraction of relevant information from the convolutional feature components, thereby enhancing the representation of critical features.

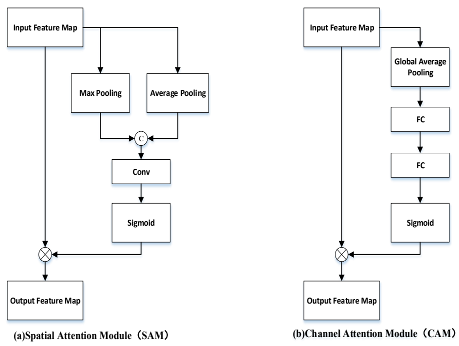

Fig. 3. The Spatial Attention (SA) module and Channel Attention (CA) module.

C. 3D MOT

The outcomes of the detection process are transmitted to the 3D multi-target tracking module, where object prediction and trajectory generation take place. In autopilot scenarios, objects exhibit diverse directions and speeds of movement. To achieve more accurate state estimation for the objects, we employ the Interacting Multiple Model (IMM) approach to combine multiple motion models. Specifically, the Constant Velocity (CV) model is utilized to estimate the object's linear motion, while the Constant Turn Rate and Velocity (CTRV) model handles the rotational motion, considering its nonlinear properties. As target occlusion can lead to changes in observed target motion patterns, this study incorporates a fusion of state estimates from various node filters to synthesize a globally consistent estimate of the target state.

1) Input Interaction and State Prediction

The IMM-UKF algorithm integrates data from diverse sensors while accounting for the uncertainty associated with each sensor's measurements. This integration is achieved by employing IMM approach, which encompasses various motion models capturing the different behaviors and dynamics of the targets. The UKF is then utilized to predict the state of each target based on the fused data.

To begin the process, data is collected from multiple sensors, such as cameras, lidars, and radars, capturing comprehensive information about the targets in the environment. To ensure consistency and alignment, all measurements are preprocessed and transformed into a common coordinate system.Next, initial probabilities are assigned to each motion model based on either prior knowledge or historical data.

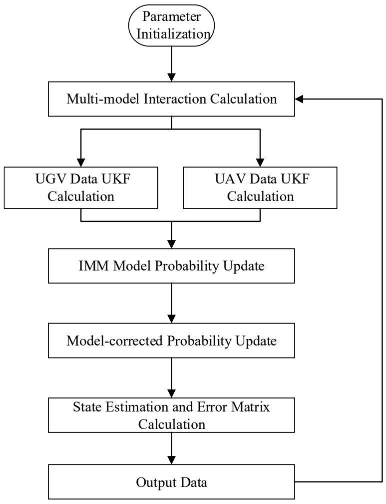

Fig. 4. The block diagram of distributed IMM-UKF algorithmic module.

2) State Update

The node's target state at the next cycle moment is updated by applying observations from neighboring nodes to calculate the posterior probability. The mode transition probabilities, governing the likelihood of switching from one motion model to another, are then updated using Bayesian filtering techniques. Additionally, the mixing probabilities, indicating the contribution of each individual Kalman filter's estimate to the final combined estimate, are updated based on the mode probabilities.

$$p^j(t)=\frac{\Lambda^j(t)\sum_{i=1}^{N}\Pi_{ij}(t-1)p^i(t-1)}{\sum_{i=1}^{N}\Lambda^i(t)\sum_{i=1}^{N}\Pi_{ij}(t-1)p^i(t-1)} \quad (6)$$

3) Data Association

To address the data association challenge in multi-target tracking, we adopt the Minimum-update Successive Shortest Path method. This algorithm leverages the 3D joint intersection-over-union to quantify the similarity between predicted and detected results, effectively establishing the best matching correspondence. The objective is to optimize data association by considering measurement uncertainties and minimizing the total association cost. The cost matrix, usually based on the distance between measurements and predicted target states, plays a crucial role in this process. By minimizing the total association cost while adhering to constraints like one measurement being assigned to exactly one target, the MUSP algorithm ensures robust associations.

$$\mathrm{C}_{\text{total}}=\sum_{i\in M}\sum_{j\in T}c_{ij}\cdot X_{ij} \quad (7)$$

subject to:

$$\sum_{j\in T}X_{ij}=1 \quad , \quad \forall i\in M \quad (8)$$

$$\sum_{i\in M}X_{ij}<1 \quad , \quad \forall j\in T \quad (9)$$

In evaluating the multi-target tracking algorithm's performance, we employ metrics such as ID switching rate, tracking accuracy, precision, and recall. Moreover, fine-tuning IMM-UKF parameters, including motion model probabilities and covariance matrices, enables us to achieve optimal performance and adaptability in various tracking scenarios.

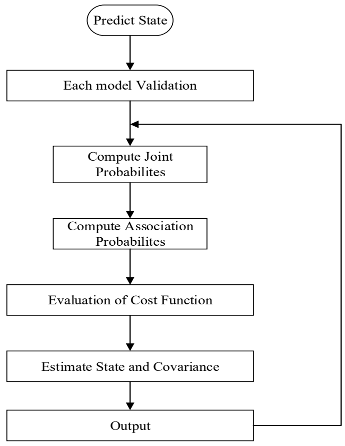

Fig. 5. The update flowchart of filter module algorithm.

IV. RESULTS

A. *Vehicle Tracking Scenarios*

In our experimental scenario, we focus on a loop turn scenario where our self-vehicle remains stationary while the target vehicle performs an irregular loop motion. To evaluate the performance of our tracking method, we compare the IMM-UKF tracking results before and after implementing state update and data association improvements. As depicted in Fig. 6, our enhanced approach achieves smoother motion trajectories and demonstrates enhanced robustness in estimating the target's speed and heading. Additionally, Fig. 6 illustrates the accurate estimation of the target's motion model, identifying it as a CTRV model, which aligns well with the actual state.

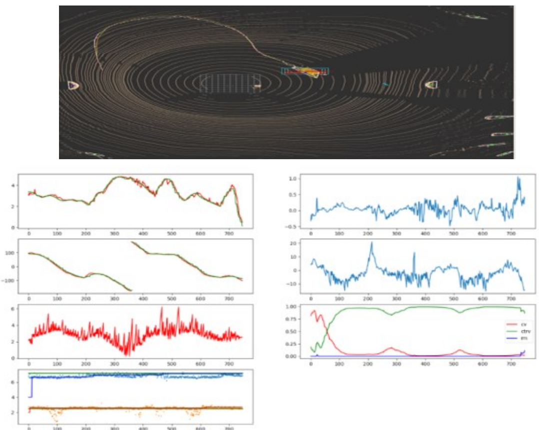

Fig. 6. Comparison results in terms of trajectory, speed, heading, size and the probabilities of the motion model.

A notable distinction exists between the two tracking methods concerning the accuracy of position, velocity, and heading estimation. This discrepancy arises from a scenario where a public transportation vehicle obstructs the tracked target vehicle. In tackling this challenge, our proposed method effectively addresses the issue of substantial errors caused by target loss in such scenarios. As illustrated in the figure, at the horizontal coordinate time of 800, the original method exhibits significant errors in speed and distance curves due to the influence of the bus stopping on the right side, with speed error reaching 2m/s and distance error reaching 2.5m. In contrast, the new method efficiently tracks the target, reducing the distance error to less than 0.5m, and maintaining speed curves closely aligned with the actual values.

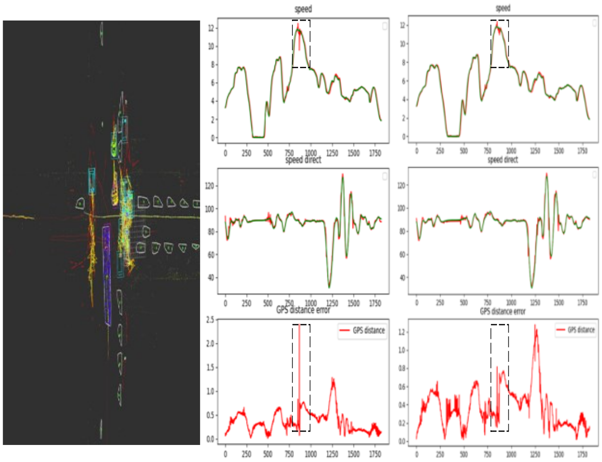

Fig. 7. Comparison of errors in speed, GPS distance, and direction between the proposed method and IMM-UKF method in occlusion scenarios.

B. *Vehicles Obscuring Pedestrian Scenarios*

We conducted cooperative tracking experiments using UAVs and drones in real road scenarios. In this setup, UAVs captured images and LiDAR data from vehicle viewpoints, while drones acquired image information from overhead viewpoints. The goal was to track target vehicles and pedestrians with multiple targets. The detection and tracking results were separately obtained for each unmanned vehicle and unmanned aircraft. Additionally, the tracking results from the fusion of the two datasets were also evaluated.

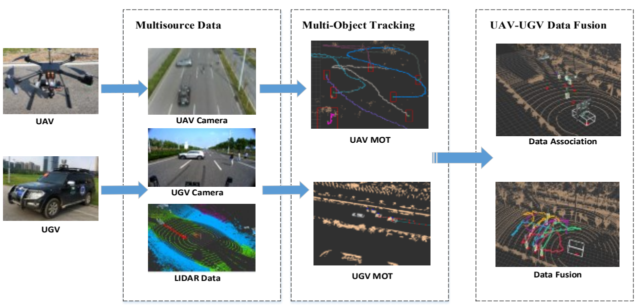

Fig. 8. UGV-UAV 3D multi-target cooperative tracking workflow in mixed pedestrian-vehicle scenarios.

We employed UAV and unmanned vehicle cooperative tracking to address scenarios where pedestrians are occluded by vehicles. In such cases, the target is lost in the visual image and LiDAR data of the unmanned vehicle. However, the UAV's visual image can still detect and track the target. Experimental results demonstrate that our proposed multi-source data fusion tracking method effectively maintains accurate estimations of the occluded target's motion pattern. Notably, the error in the target's motion trajectory behind the vehicle, as well as the state values like motion speed, remains minimal. As a result, the overall robustness and reliability of the tracking system are significantly improved.

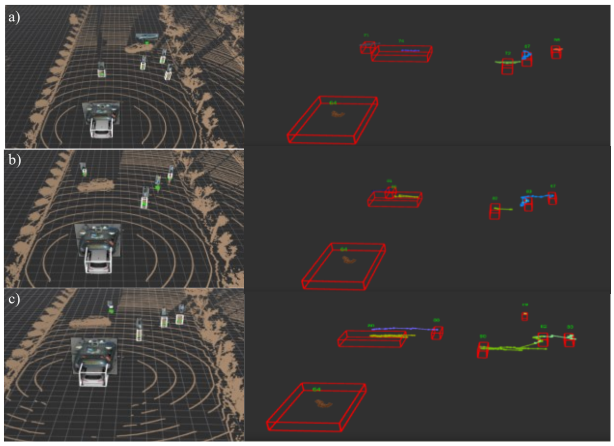

Fig. 9. Sequential frames in a multi-target tracking scene showing continuity of tracked target trajectories.

In the experiment, the motion trajectories of pedestrians were continuously tracked to prevent target loss resulting from occlusion. The evaluation revealed a substantial enhancement in ID switching, demonstrating the effectiveness of the IMM-UKF method in maintaining robust and consistent multi-target tracking. This improvement is particularly noteworthy in scenarios with vehicle-obscured pedestrians, where challenges such as occlusions and object interactions are prevalent. Moreover, the observed improvement emphasizes the potential of the IMM-UKF method for real-world applications, where accurate and reliable tracking of multiple targets plays a vital role in informed decision-making and situational awareness.

V. CONCLUSION

In this paper, We propose a UGV-UAV cooperative tracking framework designed to address target occlusion scenarios, which combines 3D object detection with 3D multi-target tracking. The 3D object detection utilizes a fusion of LiDAR point cloud and RGB images from multiple sources. Based on these detection results, the 3D multi-target object tracking employs an improved distributed IMM-UKF algorithm to estimate the states of the objects. This framework achieves effective tracking of objects and demonstrates particularly promising results in tracking tests involving target occlusion scenarios. Specifically, the tracking accuracy excels in vehicle occlusion scenarios, while the ID switching problem is successfully resolved in pedestrian occlusion scenarios.

In future work, we aim to enhance the speed of the system while maintaining detection accuracy. Alternatively, we will explore the possibility of conducting further research to realize end-to-end detection and tracking mechanisms.

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